

Smooth Regularization and Optimal Control of a Tumor–Immune–Chemotherapy Model

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Abstract

This project investigates a nonlinear ordinary differential equation model describing interactions among tumor cells, immune effector cells, and chemotherapy drug concentration, with the goal of understanding its qualitative dynamics and treatment-related control effects. The original model contains Heaviside threshold functions representing chemotherapy activation and drug-induced immune toxicity, which make the vector field nonsmooth at treatment thresholds. To address this, we introduced a smooth sigmoid regularization of the threshold terms and analyzed the resulting system using classical ODE methods. The regularized model was shown to admit locally unique solutions and to preserve biological admissibility through positivity and forward invariance of the nonnegative orthant. We also characterized equilibrium behavior, derived a local stability condition for the tumor-free equilibrium, and connected this stability analysis to treatment design through an optimal control framework. Using Pontryagin’s Maximum Principle, we derived optimal chemotherapy and immune-cell infusion controls and implemented numerical simulations comparing no treatment, constant treatment, and optimal treatment schedules. The results showed that optimal time-dependent treatment produced the lowest tumor burden among the three scenarios and reduced the tumor population to a near-zero level by the end of the treatment horizon. Overall, this project demonstrates how smooth regularization links biologically motivated threshold models with classical dynamical systems analysis and optimal control methods in mathematical oncology.

1 Introduction

Mathematical models of tumor–immune interactions have become an essential tool for understanding cancer progression and for evaluating therapeutic strategies *in silico*. These models capture the complex interplay between tumor growth, immune surveillance, and treatment-induced cytotoxicity, often through nonlinear dynamical systems that incorporate feedback regulation, saturation kinetics, and immune exhaustion [1, 2]. A recurring feature in pharmacokinetic–pharmacodynamic (PK/PD) modeling is the use of threshold-based activation of drug effects, where cytotoxicity or immune suppression is assumed to occur only when the drug concentration exceeds a biologically meaningful level [3, 4]. While such threshold formulations are biologically motivated, they introduce discontinuities into the governing equations and thereby place the system outside the scope of classical smooth ODE theory.

Nonsmooth dynamical systems of this type pose analytical challenges: the presence of Heaviside functions prevents direct application of standard results on existence and uniqueness, obstructs Jacobian-based linearization, and complicates stability analysis near switching surfaces [5, 6]. These difficulties motivate the use of smooth regularization techniques, in which discontinuous switching terms are replaced by steep but differentiable sigmoid functions. Such regularizations preserve the qualitative interpretation of threshold behavior while producing a continuously differentiable vector field amenable to classical analysis [7]. In the context of tumor–immune modeling, smooth approximations have been shown to retain biologically relevant dynamics while enabling rigorous mathematical treatment of equilibria, invariant sets, and local stability [8].

In this project, we study a tumor–immune–chemotherapy model that incorporates two threshold-dependent treatment mechanisms: drug-induced tumor cell killing and drug-induced immune-cell toxicity. To overcome the analytical limitations imposed by the Heaviside terms, we introduce a smooth sigmoid regularization and analyze the resulting system using tools from classical ODE theory. Our goals are threefold: (i) to establish fundamental qualitative properties such as positivity and forward invariance of solutions; (ii) to characterize equilibrium states and determine their local stability; and (iii) to connect these dynamical insights to treatment design through an optimal control framework. By formulating chemotherapy infusion and immune-cell administration as control variables, we apply Pontryagin’s Maximum Principle to derive optimal treatment strategies and compare them with constant or no-treatment scenarios.

This work illustrates how smooth regularization provides a mathematically tractable bridge between biologically motivated switching models and the analytical machinery of classical dynamical systems. The resulting framework enables both rigorous qualitative analysis and computational exploration of treatment optimization in mathematical oncology.

2 Original Model

We consider a nonlinear ordinary differential equation model describing the interaction among tumor cells, immune effector cells, and chemotherapy drug concentration. The model is adapted from a tumor-immune interaction framework combined with a pharmacokinetic/pharmacodynamic description of doxorubicin treatment. In this system, $E(t)$ denotes the number of immune effector cells, $T(t)$ denotes the number of tumor cells, and $M(t)$ denotes the concentration of doxorubicin at time t .

The model is given by

$$\frac{dE}{dt} = s + \frac{pET}{g+T} - mET - d_1E + s_1 - J(M)E, \quad (1)$$

$$J(M) = K_I(1 - e^{-M})H_1(M - M_{\max}), \quad (2)$$

$$\frac{dT}{dt} = aT(1 - bT) - nET - L(T, M), \quad (3)$$

$$L(T, M) = k(M - M_{th})H(M - M_{th})T, \quad (4)$$

$$\frac{dM}{dt} = u - \gamma M. \quad (5)$$

Here H and H_1 are Heaviside threshold functions which model the onset of chemotherapy effects. In particular,

$$H(M - M_{th}) = \begin{cases} 0, & M < M_{th}, \\ 1, & M \geq M_{th}, \end{cases} \quad H_1(M - M_{\max}) = \begin{cases} 0, & M < M_{\max}, \\ 1, & M \geq M_{\max}. \end{cases}$$

Thus, the drug is assumed to have a significant anti-tumor effect only when its concentration exceeds the threshold M_{th} , while toxicity to effector cells becomes significant only when the concentration exceeds the tolerable level $M_{\max}$.

The terms in (1) and (3) have natural biological interpretations. The constant term s represents the natural influx of effector cells to the tumor site. The term $\frac{pET}{g+T}$ models stimulation of effector cells by tumor presence and follows Michaelis–Menten type kinetics. The terms mET and d_1E account for effector cell exhaustion and natural death, respectively, while s_1 represents externally administered anti-tumor effector cells. In the tumor equation, the term $aT(1 - bT)$ describes logistic tumor growth with intrinsic growth rate a and carrying capacity $1/b$, and the term nET represents tumor cell destruction by effector cells. Finally, (5) describes the pharmacokinetics of doxorubicin, where u is the drug input rate and γ is the decay rate.

This model is attractive for our purposes because it incorporates nonlinear interaction terms, threshold-based treatment effects, and biologically meaningful control variables such as treatment input and timing. At the same time, the use of Heaviside functions introduces nonsmoothness into the vector field at the treatment thresholds. This presents a difficulty for applying standard ODE tools such as existence and uniqueness theorems, Jacobian linearization, and local stability analysis directly at the threshold values.

3 Smooth Regularization

To make the model more accessible to the analytical methods developed in this course, we introduce a smooth regularization of the threshold terms. The original system uses Heaviside functions to represent abrupt switching of chemotherapy effects at the thresholds M_{th} and $M_{\max}$. While this is a reasonable modeling idealization, it produces a nonsmooth vector field and complicates the direct use of standard existence, uniqueness, and linearization results from classical ODE theory.

A standard way to regularize a threshold function is to replace the Heaviside step by a smooth sigmoid. For $\varepsilon > 0$, define

$$H_\varepsilon(x) = \frac{1}{1 + e^{-x/\varepsilon}}.$$

Then $H_\varepsilon(x)$ is smooth for all x , and for small ε it approximates the step-like behavior of the Heaviside function. Using this approximation, we define the regularized treatment terms

$$J_\varepsilon(M) = K_I(1 - e^{-M})H_\varepsilon(M - M_{\max}), \quad (6)$$

$$L_\varepsilon(T, M) = k(M - M_{th})H_\varepsilon(M - M_{th})T. \quad (7)$$

The regularized tumor–immune–drug model is therefore

$$\frac{dE}{dt} = s + \frac{pET}{g+T} - mET - d_1E + s_1 - J_\varepsilon(M)E, \quad (8)$$

$$\frac{dT}{dt} = aT(1 - bT) - nET - L_\varepsilon(T, M), \quad (9)$$

$$\frac{dM}{dt} = u - \gamma M. \quad (10)$$

The purpose of this regularization is not to replace the original biological model, but rather to provide a mathematically tractable approximation that preserves its main qualitative features. In the original formulation, the drug effect is assumed to turn on abruptly once the concentration crosses a threshold. In the regularized model, this abrupt switch is replaced by a steep but continuous transition. Hence the interpretation of threshold behavior is preserved at a qualitative level, while the system becomes smooth enough for the application of standard ODE methods.

This regularization is particularly useful for several reasons. First, the right-hand side of the system becomes continuously differentiable on the biologically relevant region, so standard local existence and uniqueness results apply. Second, one may compute the Jacobian matrix of the vector field and carry out local stability analysis of equilibrium points in the usual way. Third, the regularized system remains closely connected to the original biological mechanism, since small values of ε yield dynamics that approximate threshold-type activation.

In what follows, we will analyze the regularized system (8)–(10). This allows us to study positivity of solutions, equilibrium configurations, and local stability properties using the framework of classical ordinary differential equations, while still retaining the essential treatment-threshold structure of the original model.

4 Analysis

4.1 Local existence and uniqueness

We first note that the right-hand side of (8)–(10) is smooth on the biologically relevant region

$$\Omega = \{(E, T, M) \in \mathbb{R}^3 : T > -g\}.$$

Indeed, the sigmoid H_ε is smooth for all real arguments, and the only denominator appearing in the system is $g + T$. Since the parameter g is positive in the original model, the denominator does not vanish on Ω . The remaining terms are compositions of smooth algebraic and exponential functions. Therefore the vector field is continuously differentiable on Ω , and hence locally Lipschitz there.

By the Picard–Lindelöf theorem, it follows that for every initial condition $(E_0, T_0, M_0) \in \Omega$, there exists a unique local solution to the regularized system. In particular, for biologically meaningful initial conditions with $E_0, T_0, M_0 \geq 0$, the regularized system admits a unique local solution.

4.2 Positivity and forward invariance

Theorem 1. *Assume that all model parameters are nonnegative and that the treatment input satisfies $u(t) \geq 0$ and $s_1(t) \geq 0$ for all $t \geq 0$. If*

$$E(0) \geq 0, \quad T(0) \geq 0, \quad M(0) \geq 0,$$

then every solution of the regularized system remains in the nonnegative orthant for all future time.

Proof. We check the direction of the vector field on each boundary plane of the nonnegative orthant.

First, suppose that $E = 0$ while $T \geq 0$ and $M \geq 0$. Then,

$$\frac{dE}{dt} = s + \frac{p(0)T}{g+T} - m(0)T - d_1(0) + s_1 - J_\varepsilon(M)(0) = s + s_1 \geq 0.$$

Thus the vector field points inward, or at worst tangent to the boundary, on the plane $E = 0$.

Next, suppose that $T = 0$ while $E \geq 0$ and $M \geq 0$. Then,

$$\frac{dT}{dt} = a(0)(1 - b \cdot 0) - nE(0) - L_\varepsilon(0, M).$$

Since $L_\varepsilon(T, M)$ contains a factor of T , we have $L_\varepsilon(0, M) = 0$, and therefore

$$\frac{dT}{dt} = 0.$$

Hence the vector field is tangent to the boundary plane $T = 0$ and does not point into the region $T < 0$.

Finally, suppose that $M = 0$ while $E \geq 0$ and $T \geq 0$. Then,

$$\frac{dM}{dt} = u \geq 0.$$

So again the vector field points inward or is tangent on the boundary plane $M = 0$.

Since the vector field never points outward across any boundary plane of the nonnegative orthant, a solution starting in

$$\mathbb{R}_{\geq 0}^3 = \{(E, T, M) : E \geq 0, T \geq 0, M \geq 0\}$$

cannot leave this set. Therefore the nonnegative orthant is forward invariant under the regularized dynamics. $\square$

The theorem shows that the regularized model preserves biological admissibility: negative cell populations and negative drug concentrations cannot arise from nonnegative initial data.

4.3 Linearization and local stability

Theorem 2. *Let X^* be an equilibrium of the regularized system, and assume that X^* is hyperbolic; that is, no eigenvalue of $DF(X^*)$ has zero real part. Then:*

- *if all eigenvalues of $DF(X^*)$ have negative real part, X^* is locally asymptotically stable;*
- *if at least one eigenvalue of $DF(X^*)$ has positive real part, X^* is unstable.*

Let

$$X(t) = \begin{pmatrix} E(t) \\ T(t) \\ M(t) \end{pmatrix}, \quad F(X) = \begin{pmatrix} f_1(E, T, M) \\ f_2(E, T, M) \\ f_3(E, T, M) \end{pmatrix},$$

where

$$\begin{aligned} f_1(E, T, M) &= s + \frac{pET}{g+T} - mET - d_1E + s_1 - J_\varepsilon(M)E, \\ f_2(E, T, M) &= aT(1-bT) - nET - L_\varepsilon(T, M), \\ f_3(E, T, M) &= u - M. \end{aligned}$$

Then the regularized model may be written compactly as

$$\dot{X} = F(X).$$

An equilibrium point $X^* = (E^*, T^*, M^*)^T$ is a point satisfying

$$F(X^*) = 0.$$

Then

$$DF(E, T, M) = \begin{pmatrix} \frac{pT}{g+T} - mT - d_1 - J_\varepsilon(M) & \frac{pgE}{(g+T)^2} - mE & -E J'_\varepsilon(M) \\ -nT & a - 2abT - nE - \frac{\partial L_\varepsilon}{\partial T}(T, M) & -\frac{\partial L_\varepsilon}{\partial M}(T, M) \\ 0 & 0 & -1 \end{pmatrix}.$$

Evaluating this matrix at an equilibrium X^* yields the linearization matrix

$$A := DF(E^*, T^*, M^*).$$

To derive the linearized system, let

$$Y(t) = X(t) - X^*.$$

Then

$$\dot{Y} = F(X^* + Y) - F(X^*).$$

Since $F(X^*) = 0$ and F is differentiable, Taylor's theorem gives

$$F(X^* + Y) = F(X^*) + DF(X^*)Y + R(Y),$$

where the remainder satisfies

$$\frac{\|R(Y)\|}{\|Y\|} \rightarrow 0 \quad \text{as } Y \rightarrow 0.$$

By the theorem above, local stability is determined by the eigenvalues of the Jacobian matrix

$$A = DF(E^*, T^*, M^*).$$

We therefore compute the spectrum of A . For the present model, one eigenvalue is immediately visible from the third equation, since the third row of the Jacobian is $(0, 0, -1)$. In particular, the drug component contributes a stable direction with eigenvalue -1 . For the remaining eigen values,

$$-E^* J'_\varepsilon(M^*), \quad -\frac{\partial L_\varepsilon}{\partial M}(T^*, M^*).$$

To make these terms explicit, note that

$$J_\varepsilon(M) = K_I(1 - e^{-M})H_\varepsilon(M - M_{\max}),$$

so by the product rule,

$$J'_\varepsilon(M) = K_I e^{-M} H_\varepsilon(M - M_{\max}) + K_I(1 - e^{-M})H'_\varepsilon(M - M_{\max}).$$

Similarly,

$$L_\varepsilon(T, M) = k(M - M_{th})H_\varepsilon(M - M_{th})T,$$

and therefore

$$\frac{\partial L_\varepsilon}{\partial T}(T, M) = k(M - M_{th})H_\varepsilon(M - M_{th}),$$

while

$$\frac{\partial L_\varepsilon}{\partial M}(T, M) = kT [H_\varepsilon(M - M_{th}) + (M - M_{th})H'_\varepsilon(M - M_{th})].$$

Hence

$$\det(\lambda I - A) = (\lambda + 1) [(\lambda - a_{11})(\lambda - a_{22}) - a_{12}a_{21}].$$

Therefore the eigenvalues are

$$\lambda_1 = -1,$$

and

$$\lambda_{2,3} = \frac{(a_{11} + a_{22}) \pm \sqrt{(a_{11} + a_{22})^2 - 4(a_{11}a_{22} - a_{12}a_{21})}}{2}.$$

In particular, if

$$a_{11} + a_{22} < 0 \quad \text{and} \quad a_{11}a_{22} - a_{12}a_{21} > 0,$$

then all eigenvalues of A have negative real part, and hence, by the linearization theorem, the equilibrium X^* is locally asymptotically stable.

The tumor-free equilibrium

We now consider the biologically relevant special case in which the tumor population vanishes at equilibrium, i.e.,

$$T^* = 0.$$

This corresponds to a tumor-free steady state. In this case, the equilibrium equations (??)–(??) simplify considerably.

From (??), we immediately obtain

$$M^* = u.$$

Since $T^* = 0$, equation (??) is automatically satisfied because every term contains a factor of T^* . Equation (??) then reduces to

$$0 = s + s_1 - (d_1 + J_\varepsilon(M^*))E^*.$$

Substituting $M^* = u$, we obtain

$$E^* = \frac{s + s_1}{d_1 + J_\varepsilon(u)},$$

provided that $d_1 + J_\varepsilon(u) > 0$. Therefore the regularized system admits the tumor-free equilibrium

$$X_{TF}^* = \left(\frac{s + s_1}{d_1 + J_\varepsilon(u)}, 0, u \right).$$

We now compute the Jacobian matrix at this equilibrium. Setting $T^* = 0$ and $M^* = u$, this becomes

$$A_{TF} = \begin{pmatrix} -d_1 - J_\varepsilon(u) & \frac{pE^*}{g} - mE^* & -E^* J'_\varepsilon(u) \\ 0 & a - nE^* - k(u - M_{th})H_\varepsilon(u - M_{th}) & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

where

$$E^* = \frac{s + s_1}{d_1 + J_\varepsilon(u)}.$$

The matrix A_{TF} is upper triangular, so its eigenvalues are the diagonal entries. Hence the eigenvalues of the linearization at the tumor-free equilibrium are

$$\begin{aligned} \lambda_1 &= -d_1 - J_\varepsilon(u), \\ \lambda_2 &= a - nE^* - k(u - M_{th})H_\varepsilon(u - M_{th}), \\ \lambda_3 &= -1. \end{aligned}$$

Substituting the expression for E^* yields

$$\lambda_2 = a - n \frac{s + s_1}{d_1 + J_\varepsilon(u)} - k(u - M_{th})H_\varepsilon(u - M_{th}).$$

We summarize this in the following proposition.

Proposition 1. *Assume that $d_1 + J_\varepsilon(u) > 0$. Then the regularized tumor-immune-drug system admits a tumor-free equilibrium*

$$X_{TF}^* = \left(\frac{s + s_1}{d_1 + J_\varepsilon(u)}, 0, u \right).$$

The eigenvalues of the Jacobian matrix at this equilibrium are

$$\lambda_1 = -d_1 - J_\varepsilon(u), \quad \lambda_2 = a - n \frac{s + s_1}{d_1 + J_\varepsilon(u)} - k(u - M_{th})H_\varepsilon(u - M_{th}), \quad \lambda_3 = -1.$$

Consequently, the tumor-free equilibrium is locally asymptotically stable if and only if

$$a - n \frac{s + s_1}{d_1 + J_\varepsilon(u)} - k(u - M_{th})H_\varepsilon(u - M_{th}) < 0.$$

Proof. The existence of the equilibrium follows from the equilibrium equations, as shown above. Evaluating the Jacobian at $T^* = 0$ and $M^* = u$ yields the upper triangular matrix A_{TF} . Its eigenvalues are therefore its diagonal entries. Since $d_1 \geq 0$ and $J_\varepsilon(u) \geq 0$, we have

$$\lambda_1 = -d_1 - J_\varepsilon(u) < 0$$

under the assumption $d_1 + J_\varepsilon(u) > 0$, and clearly

$$\lambda_3 = -1 < 0.$$

Hence all eigenvalues have negative real part if and only if $\lambda_2 < 0$, which is precisely

$$a - n \frac{s + s_1}{d_1 + J_\varepsilon(u)} - k(u - M_{th})H_\varepsilon(u - M_{th}) < 0.$$

The conclusion then follows from the linearization theorem for hyperbolic equilibria. $\square$

5 Connection to Treatment Optimization

Conducting an optimal control analysis and stability analysis of this ODE system are not independent results. They both give us complementary views of the same treatment problem. In this section, we explain this connection.

5.1 Problem Formulation

We treat $u(t) \in [0, u_{\max}]$ (drug infusion rate) and $s_1(t) \in [0, s_{1,\max}]$ (immune cell infusion rate) as controls and minimize the total cost functional on a fixed time interval $[0, t_f]$ where $t_f > 0$ is the prescribed treatment horizon

Our objective is

$$J(u, s_1) = \int_0^{t_f} \left[T(t) + \frac{B_1}{2} u(t)^2 + \frac{B_2}{2} s_1(t)^2 \right] dt$$

where we penalize cumulative tumor burden and quadratic treatment costs. We want a small amount of tumor while penalizing too much drug and too many immune cells (which have side effects and costs). These weights are $B_1, B_2 > 0$

5.2 Hamiltonian and Optimality Conditions

Following Evans' note [need to insert citation here], we introduce three costate variables $\lambda_E, \lambda_T, \lambda_M$ one for each state variable and write down the Hamiltonian. The Hamiltonian for this system is

$$\begin{aligned} \mathcal{H}(E, T, M, \lambda_E, \lambda_T, \lambda_M, u, s_1) = & T + \frac{B_1}{2} u^2 + \frac{B_2}{2} s_1^2 \\ & + \lambda_E \left(s + \frac{pET}{g+T} - mET - d_1 E + s_1 - J_\epsilon(M)E \right) \\ & + \lambda_T (aT(1-bT) - nET - L_\epsilon(T, M)) \\ & + \lambda_M (u - \gamma M). \end{aligned} \quad (11)$$

For optimal control problems, the Hamiltonian puts together the objective and the dynamics so that we can directly apply Potryagin's Maximum Principle (PMP) to have conditions on the controls and costates. The $\lambda_i(t)$ variables can be thought of as measuring "How much would the total cost change if $x_i(t)$ increased by some small amount"

Theorem 3 (Potryagin's Maximum Principle). [9, 10] *Let (u^*, s_1^*) be an optimal control pair for the problem which minimizes $J(u, s_1)$ subject to the regularized constraints. Then there exist absolutely continuous costate functions $\lambda_E, \lambda_T, \lambda_M : [0, t_f] \rightarrow \mathbb{R}$ such that the following conditions hold for almost every $t \in [0, t_f]$:*

1. **(State equations)** *The state evolves according to the ODE system under controls (u^*, s_1^*)*
2. **(Adjoint equations)** *The costates satisfy $\dot{\lambda}_i = -\frac{\partial \mathcal{H}}{\partial x_i}$ for each state variable $x_i \in \{E, T, M\}$.*
3. **(Minimization condition)** *The optimal control minimizes the Hamiltonian pointwise over the admissible set:*

$$\mathcal{H}(\dots, u^*, s_1^*) \leq \mathcal{H}(\dots, u, s_1) \quad \text{for all } (u, s_1).$$

4. **(Transversality)** *Since J has no terminal cost, we have $\lambda_E(t_f) = \lambda_T(t_f) = \lambda_M(t_f) = 0$.*

Differentiating the Hamiltonian with respect to each state variable, we get the costate equations

$$\begin{aligned} \frac{d\lambda_E}{dt} &= -\frac{\partial \mathcal{H}}{\partial E} = -\lambda_E \left(\frac{pT}{g+T} - mT - d_1 - J_\epsilon(M) \right) + \lambda_T nT \\ \frac{d\lambda_T}{dt} &= -\frac{\partial \mathcal{H}}{\partial T} = -1 - \lambda_E \left(\frac{pgE}{(g+T)^2} - mE \right) - \lambda_T (a(1-2bT) - nE - \frac{\partial L_\epsilon}{\partial T}) \\ \frac{d\lambda_M}{dt} &= -\frac{\partial \mathcal{H}}{\partial M} = \lambda_E E J'_\epsilon(M) + \lambda_T \left(\frac{\partial L_\epsilon}{\partial M} \right) + \lambda_M (\gamma) \end{aligned}$$

with the boundary conditions $\lambda_E(t_f) = \lambda_T(t_f) = \lambda_M(t_f) = 0$ from the transversality conditions. These equations above are known as the costate equations for the system.

5.3 Optimal Controls

Applying the minimization condition from [9] in the Potryagin's Maximum Principle, we minimize the Hamiltonian $\mathcal{H}$ over the controls u and s_1 separately. The variable u enters $\mathcal{H}$ only through the terms $\frac{B_1}{2}u^2 + \lambda_M u$ which is strictly convex in u , this can be verified by applying the definition of convexity. Setting $\frac{\partial \mathcal{H}}{\partial u} = 0$, we get

$$B_1 u + \lambda_M = 0 \implies u = -\frac{\lambda_M}{B_1}$$

Applying a similar argument to s_1 with the terms $\frac{B_2}{2}s_1^2 + \lambda_E s_1$, we get

$$B_2 s_1 + \lambda_E = 0 \implies s_1 = -\frac{\lambda_E}{B_2}$$

We found the unconstrained minimizers for u and s_1 above, but the controls are not free to be any values. The controls need to satisfy $u \in [0, u_{\max}]$ and $s_1 \in [0, s_{1,\max}]$, so we need to project the unconstrained minimizers to this allowed range

Projecting the values onto this admissible set, we get the explicit optimal controls

$$u^*(t) = \min(u_{\max}, \max(0, -\frac{\lambda_M}{B_1})), \quad s_1^*(t) = \min(s_{1,\max}, \max(0, -\frac{\lambda_E}{B_2}))$$

The min and max enforce box constraints for us. If the unconstrained values are negative, we use 0 and if it exceeds the upper bound of the admissible set, we saturate at the maximum. Biologically, we can interpret this as drug infusion being active when $\lambda_M(t) < 0$ and immune cell infusion being active when $\lambda_E(t) < 0$

5.4 Numerical Simulations

We solve the optimal control problem above formed by the state and costate equations numerically using the forward-backward sweep method [11]. The algorithm works as follows:

- (i) Start from an initial guess for the controls
- (ii) Integrate the state equations forward from $t = 0$
- (iii) Use the resulting trajectory to integrate the costate equations backward from $\lambda_i(t_f) = 0$
- (iv) Update the controls using the optimal control formula and repeat until convergence

We compare three scenarios over a treatment period of $t_f = 30$ days. The scenarios we consider are: no treatment (where $u = 0, s_1 = 0$), constant treatment (where $u = 1.5, s = 0.2$), and the computed optimal schedule.

The three figures below show the evolution of the effector cell population $E(t)$, tumor cell population $T(t)$, and drug concentration $M(t)$ under the three scenarios. Without treatment, the tumor grows logistically toward its carrying capacity. Constant treatment suppresses tumor growth significantly. The optimal schedule drives the tumor to near-elimination by day 30, achieving the lowest final tumor of the three scenarios

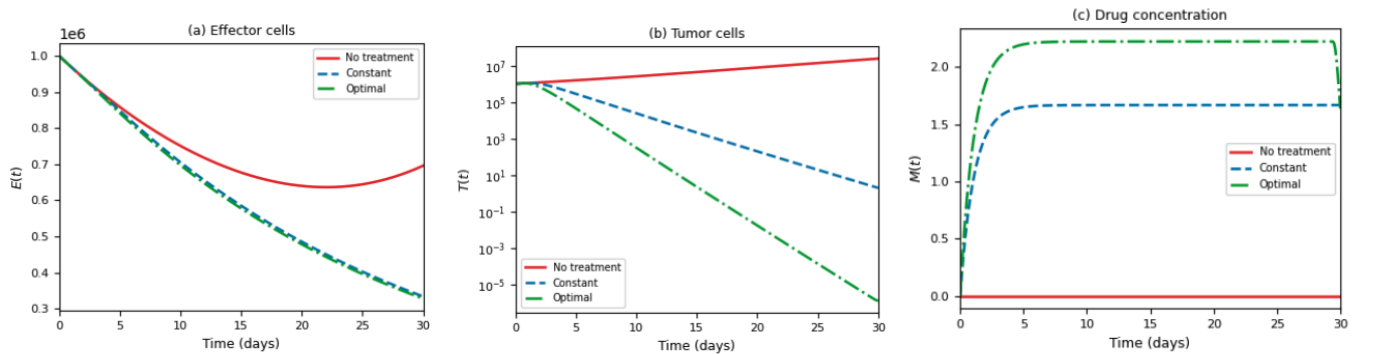


Figure 1: We have the state trajectories over $t_f = 30$ days for no treatment (red), constant treatment (blue dashed), and optimal control (green dash-dot)

The figures below show the optimal drug infusion $u^*(t)$ and immune cell infusion $s_1^*(t)$ over the treatment horizon. We can see that both controls are most intense early in the treatment period when the tumor burden is the highest, and it tapers off as the tumor gets suppressed. This is consistent with the minimization conditions: controls are large when the corresponding costates are the most negative which means the treatment is most cost-effective early.

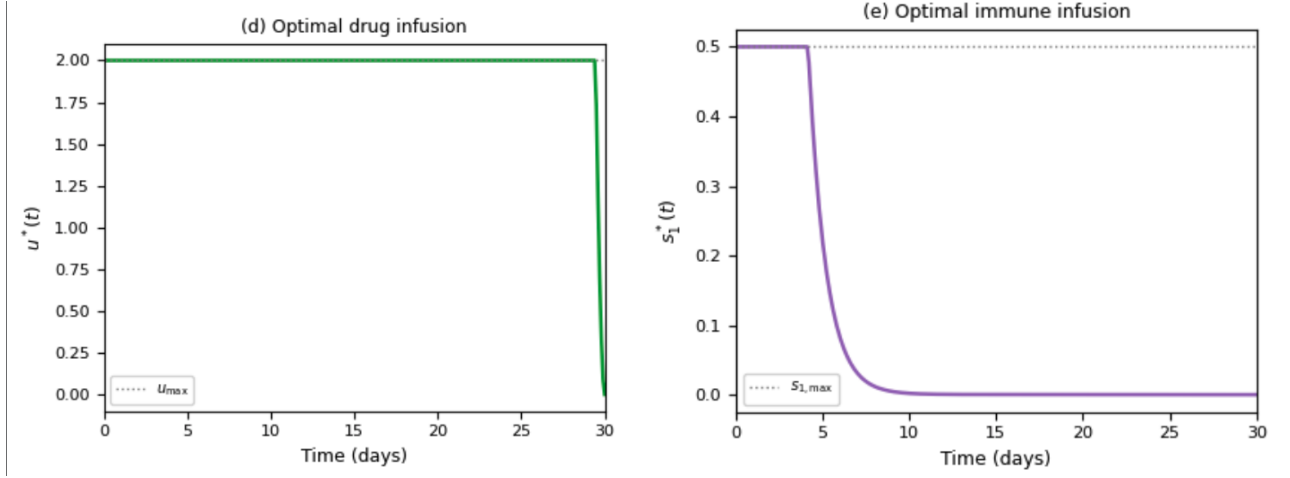


Figure 2: Optimal control inputs over $t_f = 30$ days. Left: drug infusion rate $u^*(t)$. Right: immune cell infusion rate $s_1^*(t)$. The dotted lines show the upper bounds $u_{\max}$ and $s_{1,\max}$.

The figure below shows the three costate variables $\lambda_E(t), \lambda_T(t), \lambda_M(t)$. All three costate variables satisfy $\lambda_i(t_f) = 0$, verifying the transversality conditions from Potryagin's Maximum Principle. We also have for all costate variables that $\lambda_M(t) < 0$ throughout the active treatment phase, which is precisely the condition under which drug infusion is optimal according to the formula from section 5.3.

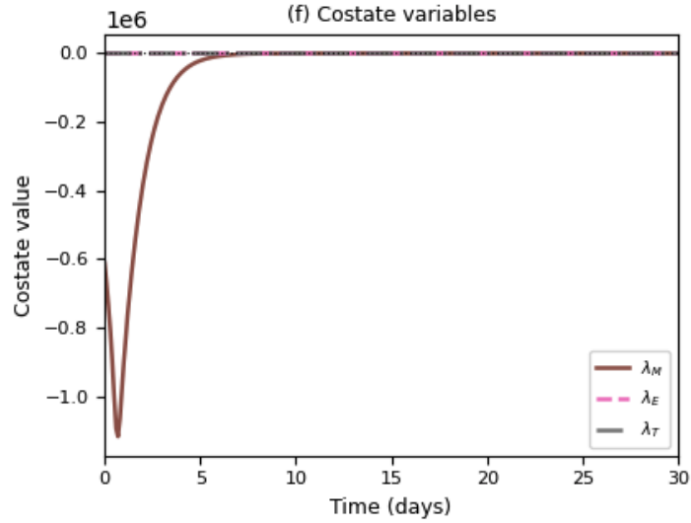


Figure 3: Costate Variables

6 Discussion

The analysis presented in this project demonstrates how smooth regularization of discontinuous treatment terms enables a rigorous mathematical study of tumor-immune-chemotherapy models that would otherwise fall outside the scope of classical ODE theory. By replacing the Heaviside threshold functions with steep sigmoid approximations, we obtained a continuously differentiable vector field while preserving the biological interpretation of abrupt treatment activation. This modification allowed us to establish local existence and uniqueness of solutions, prove positivity and forward invariance of the biologically relevant region, and characterize equilibrium states using standard linearization techniques.

A central outcome of the analysis is the explicit stability condition for the tumor-free equilibrium. The eigenvalue associated with tumor dynamics depends on the balance between intrinsic tumor growth, immune-mediated killing, and chemotherapy-induced cytotoxicity. In particular, the condition

$$a - n \frac{s + s_1}{d_1 + J_\varepsilon(u)} - k(u - M_{th})H_\varepsilon(u - M_{th}) < 0$$

highlights the competing influences of natural tumor proliferation, immune response strength, and drug concentration. This inequality provides a mathematically transparent criterion for treatment success: the combined immune and chemotherapeutic effects must outweigh the tumor's intrinsic growth rate. Such stability thresholds are consistent with classical results in tumor-immune modeling [2, 8] and offer insight into how treatment parameters influence long-term outcomes.

The optimal control results complement the stability analysis by identifying treatment schedules that minimize tumor burden while penalizing excessive drug or immune-cell administration. The structure of the optimal controls, derived from Pontryagin's Maximum Principle, reveals that both chemotherapy infusion and immune-cell supplementation are most effective when applied aggressively early in the treatment horizon. This behavior is driven by the costate dynamics, which quantify the marginal value of reducing tumor burden at each point in time. When the tumor population is large, the costates become strongly negative, leading to control saturation at the upper bounds. As the tumor is suppressed, the marginal benefit of treatment decreases, and the optimal controls taper accordingly. This pattern is consistent with findings in optimal chemotherapy scheduling [1, 4].

Numerical simulations further illustrate the interplay between system dynamics and optimal control. In the absence of treatment, the tumor grows logistically toward its carrying capacity, while constant treatment produces partial suppression but does not eliminate the tumor within the prescribed time horizon. In contrast, the optimal control strategy drives the tumor population close to zero by the end of the treatment period, achieving a substantially lower tumor burden than either no treatment or constant dosing. These results underscore the importance of time-dependent treatment strategies and demonstrate how optimal control can exploit nonlinear interactions between tumor cells, immune effectors, and drug concentration.

More broadly, this project highlights the value of smooth regularization as a bridge between biologically motivated switching models and the analytical tools of classical dynamical systems. While the original discontinuous model captures important biological thresholds, its nonsmooth structure complicates both theoretical analysis and numerical optimization. The regularized model retains the essential biological features while enabling a unified framework for studying qualitative dynamics, stability, and optimal treatment design. This approach may be extended to more complex models incorporating additional immune components, drug resistance mechanisms, or multi-drug therapies, and it provides a foundation for future work integrating data-driven parameter estimation or uncertainty quantification.

The combination of smooth regularization, dynamical systems analysis, and optimal control offers a powerful methodology for exploring treatment strategies in mathematical oncology. The results obtained here demonstrate how mathematical structure can inform therapeutic design and provide insight into the conditions under which tumor elimination is achievable.

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